03008/2&1 BECE
June 2022
MATHEMATICS 2&1
Essay and Objective
2 hours

2&1

Name.....
Index Number.....

THE WEST AFRICAN EXAMINATIONS COUNCIL

GHANA

Basic Education Certificate Examination

June 2022

MATHEMATICS 2 & 1
Essay and Objective

2 hours

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions. Write your* **name** *and* **index number** *in the spaces provided above.*

This booklet consists of two papers. Answer Paper 2 *which comes first, in your answer booklet and Paper* 1 *on your Objective Test answer sheet. Paper* 2 *will last* **1** hour *after which the answer booklet will be collected. Do* **not** *start Paper* 1 *until you are told to do so. Paper* 1 *will last* **1** hour.

The use of calculators is **not** *allowed.*

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1 hour

PAPER 2

Essay

Answer **four** *questions* **only.**

All questions carry equal marks.

All working must be clearly shown. Marks will **not** *be awarded for correct answers without corresponding working.*

1. (*a*) Given that P = {factors of 36} and Q = {factors of 54},

(i) list the members in the sets P and Q.

(ii) Find:

(α) $P \cap Q$;

(β) $n(P \cap Q)$;

(γ) the Highest Common Factor (HCF) of 36 and 54.

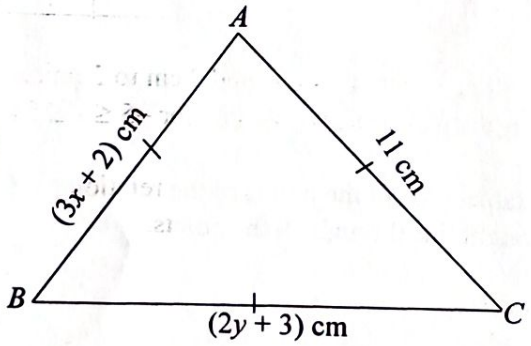
(*b*) Write down the next two terms of the sequence 1, 4, 9, . . ., ...

(*c*) The median of the ordered set of observations 2, 3, $(4m - 3)$, $(3m + 1)$, 11 and 13 in ascending order is 6. Find the value of m.

2. (*a*) Simplify : $\left(\frac{1}{3} + \frac{1}{12}\right) \div \left(\frac{2}{3} - \frac{5}{8}\right)$.

(*b*) Find the product of $(2x - 3)$ and $(2x + 3)$.

(*c*)



NOT DRAWN TO SCALE

In the diagram, ABC is an equilateral triangle. Find the value of $(x + y)$.

3. (*a*) Given the relation $L = \frac{2(m^2 - n^2)}{4(m + n)}$:

(i) simplify L;

(ii) find the value of L when $m = 2$ and $n = 3$.

(*b*) Solve $\frac{4}{3x} = 7 - \frac{3}{x}$.

(*c*) A salesman gets a commission of $5\frac{1}{2}$% of the value of items he sells. The salesman sells 12 textbooks at GH¢ 25.00 per book, 3 scientific calculators at GH¢ 50.00 per calculator and 8 packets of bic pens at GH¢ 50.00 per packet. Calculate the salesman's commission.

4. (*a*) Fred is $(x - 1)$ years old now. How old:
   (i) was he 4 years ago?
   (ii) will he be 8 years from now?
   (iii) is he now, if his age in 8 years time will be three times his age 4 years ago?

   (*b*) The perimeter of a rectangular cocoa farm is 497 km. The length of the farm is $2\frac{1}{2}$ times the width. Find the:
   (i) width;
   (ii) length of the farm.

5. (*a*) Factorize: $(x - y)(3m + n) - (x - y)(m - 2n)$.

   (*b*) Given that $\mathbf{p} = \begin{pmatrix} 2 - 3x \\ 5 - 2y \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$ and $\mathbf{p} - \mathbf{q} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$, find the value of $(x + y)$.

   (*c*) (i) Find the truth set of $\frac{x - 1}{2} \leq \frac{1}{2} + x$.

   (ii) Illustrate the answer in (i) on the number line.

6. (*a*) Copy and complete the table for the relation $y = 5 - 2x$ for $-3 \leq x \leq 4$.

x	−3	−2	−1	0	1	2	3	4
y	11			5		1		−3

   (*b*) Using a scale of 2 cm to 1 unit on x-axis and 2 cm to 2 units on the y-axis, draw on a graph sheet two perpendicular axes ox and oy for $-5 \leq x \leq 5$ and $-12 \leq y \leq 12$.

   (*c*) (i) Using the table, plot all the points of the relation $y = 5 - 2x$.
   (ii) Draw a straight line through all the points.

   (*d*) Using the graph, find the:
   (i) value of y when $x = -2.6$;
   (ii) value of x when $y = -2.8$;
   (iii) gradient of the line.

END OF ESSAY TEST

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

YOU WILL BE PENALIZED SEVERELY IF YOU ARE FOUND LOOKING AT THE NEXT PAGE BEFORE YOU ARE TOLD TO DO SO.

PAPER 1 **1 hour**

OBJECTIVE TEST

Answer all the questions on your Objective Test answer sheet.

1. Use **2B** pencil throughout.
2. On the pre-printed answer sheet, check that the following details are **correctly** printed: Your **surname** followed by your **other names**, the *Subject Name*, your *Index Number*, *Centre Number* and the *Paper Code*.
3. In the boxes marked *Candidate Number, Centre Number* and *Paper Code*, **reshade** each of the shaded spaces.
4. An example is given below. This is for a female candidate whose name is Dora Afi SULE. Her *index number* is 772384188 and she is writing the examination at *Centre Number* 77234. She is offering *Mathematics* 1 and the *Paper Code is* 0301.

THE WEST AFRICAN EXAMINATIONS COUNCIL, GHANA
BASIC EDUCATION CERTIFICATE EXAMINATION
OBJECTIVE ANSWER SHEET

CANDIDATE NAME:	SUBJECT NAME:
SULE DORA AFI	MATHEMATICS 1

1. Use 2B pencil, Press firmly.
2. Answer each question by choosing one letter and then, shade through the letter chosen like this ⊏A⊐ ⊏C⊐ ⊏D⊐ ⊏E⊐
3. If you want to change an answer, erase your first mark completely.
4. If only four alternative answers are given for each question, ignore the letter E.
5. Your question paper may have fewer than 60 questions.

CANDIDATE NUMBER								
7	7	2	3	8	4	1	8	8

CENTRE NUMBER				
7	7	2	3	4

PAPER CODE			
0	3	0	1

For Supervisors only. If candidate is absent shade this space.

Answer **all** *questions.*

Each question is followed by four options lettered **A** *to* **D**. *Find the* **correct** *option for* **each** *question and shade in* **pencil** *on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

If $3n + 2 = 8$, find the value of n.

A. 10

B. 6

C. 3

D. 2

The correct answer is 2, *which is lettered* D *and therefore answer space* D *would be shaded.*

⊏A⊐ ⊏B⊐ ⊏C⊐ ⊏D⊐

Think carefully before you shade the answer spaces. Erase completely any answers you wish to change.

Do all rough work on this question paper.

Now answer the following questions.

1. Expand: $(2a - b)(a - b)$.

A. $2a^2 + 3ab + b$

B. $2a^2 - 3ab + b^2$

C. $2a^2 - 3ab - b^2$

D. $2a^2 + 3ab - b$

2. A shop is rented at GH¢ 9.00 per month. How much money is paid in $1\frac{1}{2}$ years?

A. GH¢ 6.00

B. GH¢ 13.50

C. GH¢ 135.00

D. GH¢ 162.00

3. If a trader made a profit of 10 % in selling a shirt for GH¢ 44.00, find the cost price.

A. GH¢ 38.50

B. GH¢ 39.60

C. GH¢ 40.00

D. GH¢ 48.40

4. The volume of a cylinder is 40π cm^3. If the height of the cylinder is 10 cm, find the base radius.

A. 1 cm
B. 2 cm
C. 3 cm
D. 4 cm

5. The **longest** chord of a circle is the

A. circumference.
B. diameter.
C. sector.
D. segment.

6. Arrange the following fractions from the lowest to the highest: $\frac{3}{5}, \frac{1}{4}, \frac{2}{3}$.

A. $\frac{3}{5}, \frac{1}{4}, \frac{2}{3}$
B. $\frac{1}{4}, \frac{3}{5}, \frac{2}{3}$
C. $\frac{2}{3}, \frac{1}{4}, \frac{3}{5}$
D. $\frac{2}{3}, \frac{3}{5}, \frac{1}{4}$

7. Express 108 as a product of prime factors.

A. $2^2 \times 3^3$
B. $2^3 \times 3^2$
C. 2×3^4
D. $2^3 \times 3$

8. The point $S(4, 3)$ is reflected in the y-axis. Find the coordinates of the image of S.

A. $(-3, 4)$
B. $(3, -4)$
C. $(-4, 3)$
D. $(4, -3)$

9. Factorize completely $xy - xm - my + m^2$.

A. $(x - m)(y - m)$
B. $(m - y)(x - m)$
C. $(y - m)(m - x)$
D. $(m - y)(m - x)$

10. The rule of a mapping is $x \rightarrow 2x^2 - 1$. What number does $x = 2$ map to?

A. 3
B. 7
C. 8
D. 9

11. Which of the following is the **largest** set?

A. {Composite}
B. {Integers}
C. {Natural numbers}
D. {Whole numbers}

12. What is the probability that a number greater than 5 shows up when a die is thrown?

A. $\frac{1}{6}$
B. $\frac{1}{3}$
C. $\frac{2}{3}$
D. $\frac{5}{6}$

13. Simplify: $-15-(-20)+(-10)$.

A. -45
B. -25
C. -5
D. 5

14. What is the length of the side of a square of area 225 cm^2?

A. 12.00 cm
B. 15.00 cm
C. 56.25 cm
D. 112.50 cm

15. Adwoa travelled 12 km due north and 5 km due east. How far was she from her starting point?

A. 7 km
B. 13 km
C. 17 km
D. 60 km

16. Simplify $1\frac{1}{5} \div 2\frac{1}{10}$.

A. $\frac{4}{7}$

B. $\frac{25}{63}$

C. $\frac{12}{63}$

D. $2\frac{13}{25}$

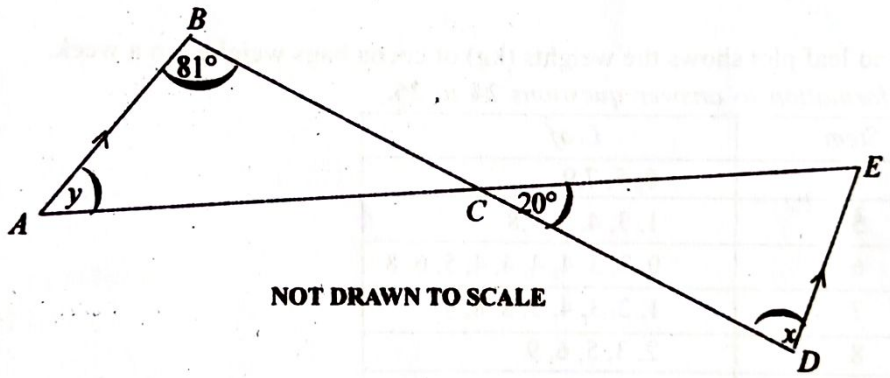
17. A car travels 36 kilometres in an hour. Find its speed in metres per second.

A. 10 m s^{-1}

B. 20 m s^{-1}

C. 100 m s^{-1}

D. 200 m s^{-1}



NOT DRAWN TO SCALE

In the diagram, *AB* is parallel to *DE*, angle *ABC* = 81° and angle *DCE* = 20°.
Use the information to answer questions **18** *and* **19**.

18. What is the value of x?

A. 20°

B. 61°

C. 81°

D. 101°

19. Find the value of y.

A. 20°

B. 81°

C. 79°

D. 101°

20. For what value of x is $3^x = 81$?

A. 2

B. 4

C. 9

D. 27

21. A football match starts at 2.20 p.m. and lasts for 1 hour 50 minutes. At what time will the game end?

A. 3.10 p.m.
B. 4.10 p.m.
C. 5.10 p.m.
D. 6.10 p.m.

22. The ratio of farmers to children in a village is 13:11. If there were 312 farmers in the village, how many children were there?

A. 48
B. 143
C. 169
D. 264

23. If $\frac{n}{100} = 10.5$, find the value of n.

A. 0.0105
B. 0.105
C. 105
D 1050

The stem and leaf plot shows the weights (kg) of cocoa bags weighed in a week.
Use the information to answer questions **24** *to* **26.**

Stem	*Leaf*
4	0, 5, 7,9
5	1, 3, 4, 5, 7,8
6	0, 2, 3, 4, 4, 4, 4, 5, 6, 8
7	1, 2, 3, 4, 5, 8, 8, 9
8	2, 3, 5, 6, 9
9	4, 5

24. How many bags of cocoa were weighed in the week?

A. 41
B. 35
C. 29
D. 6

25. What is the median weight of the bags of cocoa?

A. 61 kg
B. 62 kg
C. 65 kg
D. 68 kg

26. What is the modal weight of the bags of cocoa?

A. 60 kg
B. 64 kg
C. 65 kg
D. 68 kg

27. Write 1930.54 in standard form.

A. 1.93054×10^{3}
B. 1.93054×10^{2}
C. 1.93054×10^{-2}
D. 1.93054×10^{-3}

28. A mother has GH¢ 5.00 and gives **each** of her 3 children GH¢ 1.50 as pocket money. How much is left for her?

A. GH¢ 0.15
B. GH¢ 0.50
C. GH¢ 3.50
D. GH¢ 4.50

29. If $x = \dfrac{b^2 - 4ac}{2a}$, find x when $a = 2$, $b = -4$ and $c = -2$.

A. 0
B. 4
C. 8
D. 16

30. Express 15 : 12 in the form 1 : n.

A. 1 : 0.8
B. 1 : 1.2
C. 1 : 15
D. 1 : 12

31. Peter had GH¢ 200.00 and spent GH¢ 83.00. What percentage of the money is left?

A. 70.94 %
B. 58.50 %
C. 41.50 %
D. 29.06 %

32. Given that $117(12 + 18) = 117(15 + k)$, find the value of k.

A. -30

B. -15

C. 15

D. 30

33. If $\mathbf{a} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$, find $\mathbf{a} + \mathbf{b}$.

A. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

B. $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

C. $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

D. $\begin{pmatrix} 0 \\ 6 \end{pmatrix}$

34. Solve: $(x - 1) = \frac{1}{2}(x + 2)$.

A. 0

B. $1\frac{1}{3}$

C. $1\frac{1}{2}$

D. 4

35. Which of the following is **not** a quadrilateral?

A. Parallelogram

B. Rhombus

C. Triangle

D. Rectangle

36. Given that $\overrightarrow{PQ} = \begin{pmatrix} -3 \\ 8 \end{pmatrix}$, find $\overrightarrow{QP}$.

A. $\begin{pmatrix} -3 \\ 8 \end{pmatrix}$

B. $\begin{pmatrix} 8 \\ -3 \end{pmatrix}$

C. $\begin{pmatrix} -3 \\ -8 \end{pmatrix}$

D. $\begin{pmatrix} 3 \\ -8 \end{pmatrix}$

37. A certain number is subtracted from 12 and the result is multiplied by 3. If the answer is 21, find the number.

A. 4
B. 5
C. 6
D. 8

38. Solve: $5x - (7x - 3) \leq 9$.

A. $x \leq -3$
B. $x \geq -3$
C. $x \geq -6$
D. $x \leq 3$

39. Express $\frac{3}{8}$ as a decimal fraction.

A. 0.375
B. 0.429
C. 0.365
D. 0.625

40. Given that $P(3, -2)$ and $Q(-2, 4)$ are points in a plane, find the gradient of the line joining P to Q.

A. $-\frac{6}{5}$
B. $-\frac{5}{6}$
C. $\frac{5}{6}$
D. $\frac{6}{5}$

END OF PAPER